TUTORIAL 8

Linear Maps and Coordinates

MOTIVATION AND GUIDING IDEA

Motivation. The same information may appear as a polynomial, a matrix or a coordinate vector. A linear map compares such descriptions while preserving linear combinations. The kernel asks what is lost; the image asks what can be attained.

NOTATION AND CONVENTIONS

- $T : V \to W$ denotes a map with domain V and codomain W; $T(v) \in W$ is the image of v. The vectors $0_V, 0_W$ are the zero vectors in the corresponding spaces.
- $\ker T = \{v \in V : T(v) = 0_W\}$ is the kernel; $\operatorname{Im} T = T(V) = \{T(v) : v \in V\}$ is the image. Do not automatically identify the image with the codomain.
- $\operatorname{rank} T = \dim \operatorname{Im} T$ is the rank; $\dim \ker T$ is the nullity. The notation $V \cong W$ means that a linear isomorphism between them exists.
- $B = (v_1, \ldots, v_n)$ is an **ordered** basis. If $v = \sum_{j=1}^{n} a_j v_j$, then $[v]_B = (a_1, \ldots, a_n)^T$. The order of basis vectors determines the order of entries.
- $\mathbb{F}_{\leq n}[X]$ is the space of polynomials of degree at most n, including the zero polynomial. For $T : V \to V$, $T^2 = T \circ T$ denotes composition, not entrywise squaring.

DEFINITION 8.1

Let V, W be vector spaces over the same field $\mathbb{F}$. A map $T : V \to W$ is linear if it is well defined and satisfies

$$T(av + bw) = aT(v) + bT(w) \qquad (v, w \in V,\ a, b \in \mathbb{F}).$$

It is injective if $T(v) = T(w)$ implies $v = w$, and surjective if every $y \in W$ equals $T(v)$ for some $v \in V$. For a linear map, injectivity is equivalent to $\ker T = \{0_V\}$, and surjectivity to $\operatorname{Im} T = W$. An isomorphism is a bijective linear map; its inverse is linear. If $\dim V < \infty$, then

$$\dim V = \dim \ker T + \dim \operatorname{Im} T.$$

The coordinate map $v \mapsto [v]_B$ is an isomorphism $V \to \mathbb{F}^n$. A vector and its coordinate column are related by a choice of basis; they should not be equated without an appropriate identification.

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MOTIVATION AND GUIDING IDEA

Tutorial objectives. Recognize linear maps, compute kernels and images, and use coordinates to pass between polynomials, matrices and vectors. Distinguish a mathematical object from its representation in a chosen basis.

8.1 Linear maps

DEFINITION 8.2

Let V, W be vector spaces over the same field $\mathbb{F}$.
A map $T : V \to W$ is **linear** if, for all $v_1, v_2 \in V$ and $\lambda \in \mathbb{F}$,

1. $T(v_1 + v_2) = T(v_1) + T(v_2)$.
2. $T(\lambda v_1) = \lambda T(v_1)$.

Interpretation A linear map sends vectors from one vector space to another while preserving their linear structure. It respects vector addition and scalar multiplication.

The first condition says that adding and then applying the map gives the same result as applying the map and then adding. Thus addition commutes with applying the map.

The second condition says that multiplying by a scalar before applying the map gives the same result as multiplying the image by that scalar. In a real geometric setting, scaling an input by λ scales its image by the same factor.

NOTE

Together these properties imply that, for all $\alpha, \beta \in \mathbb{F}$ and $v_1, v_2 \in V$,

$$T(\alpha v_1 + \beta v_2) = \alpha T(v_1) + \beta T(v_2),$$

so a linear map preserves every linear combination.
Geometrically, real linear maps include scalings, rotations and reflections. They send a line to a line or a point and fix the zero vector. In particular, $T(0_V) = 0_W$ always holds.
The main consequences are:

- More generally, $T(\sum_{i=1}^{n} \lambda_i v_i) = \sum_{i=1}^{n} \lambda_i T(v_i)$.
- A necessary condition for $T : V \to W$ to be linear is $T(0_V) = 0_W$. (A map with $T(0_V) \neq 0_W$ is not linear.)

EXERCISE 8.1

Let $n \geq 1$, and define

$$V = \mathbb{R}_{\leq n}[X].$$

Please determine which of the following formulas define linear maps (first verify that each proposed map is **well defined** as a map $V \to V$):

1. $T_1 : V \to V$, given by
$$T_1(P(X)) := P(X) + P(0).$$

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EXERCISE 8.1 CONTINUED

2. $T_2 : V \to V$, given by
$$T_2(P(X)) := X \cdot P(X).$$

3. $T_3 : V \to V$, given by
$$T_3(P(X)) := P(X+1) - P(1).$$

WORKED SOLUTION

Recall that $T : V \to V$ is linear if, for all $P, Q \in V$ and $\alpha, \beta \in \mathbb{R}$,

$$T\big(\alpha P + \beta Q\big) = \alpha T(P) + \beta T(Q).$$

1. **T_1 is linear.**

   (a) Well-definedness. If $P \in V$, then $P(0) \in \mathbb{R}$ is constant, so $P(X) + P(0)$ has degree at most n. Hence $T_1(P) \in V$.

   (b) Linearity. Let $P, Q \in V$ and $\alpha, \beta \in \mathbb{R}$. Then

   $$T_1\big(\alpha P(X) + \beta Q(X)\big) = (\alpha P + \beta Q)(X) + (\alpha P + \beta Q)(0).$$

   Since

   $$(\alpha P + \beta Q)(X) = \alpha P(X) + \beta Q(X), \qquad (\alpha P + \beta Q)(0) = \alpha P(0) + \beta Q(0),$$

   we obtain

   $$T_1\big(\alpha P(X)+\beta Q(X)\big) = \alpha\big(P(X)+P(0)\big)+\beta\big(Q(X)+Q(0)\big) = \alpha T_1(P(X))+\beta T_1(Q(X)).$$

   Thus T_1 is linear.

2. **T_2 does not define a map $V \to V$, so it is not a linear operator on V.**

   A linear map must first be well defined with its stated domain and codomain. We show that this requirement fails for T_2.

   Take $P(X) = X^n \in V$. Then

   $$T_2(P(X)) = X \cdot X^n = X^{n+1} \notin \mathbb{R}_{\leq n}[X] = V.$$

   Thus T_2 does not define a map $V \to V$, and cannot be a linear operator on V.

   Remark (generalization). If instead we define

   $$\widetilde{T}_2 : \mathbb{R}[X] \to \mathbb{R}[X], \qquad \widetilde{T}_2(P(X)) := X \cdot P(X),$$

   then $\widetilde{T}_2$ is well defined and linear: for all $P, Q \in \mathbb{R}[X]$ and $\alpha, \beta \in \mathbb{R}$,

   $$\widetilde{T}_2(\alpha P + \beta Q) = X(\alpha P + \beta Q) = \alpha XP + \beta XQ = \alpha \widetilde{T}_2(P) + \beta \widetilde{T}_2(Q).$$

3. **T_3 is linear.**

   (a) Well-definedness. If $P \in V$, then $\deg P \leq n$, so $\deg P(X+1) \leq n$ as well. Also, $P(1) \in \mathbb{R}$ is constant, so $P(X+1) - P(1) \in V$. Hence $T_3(P) \in V$.

   (b) Linearity. Let $P, Q \in V$ and $\alpha, \beta \in \mathbb{R}$. Then

   $$T_3\big(\alpha P(X) + \beta Q(X)\big) = (\alpha P + \beta Q)(X+1) - (\alpha P + \beta Q)(1).$$

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EXERCISE 8.1 CONTINUED

Since

$$(\alpha P+\beta Q)(X+1)=\alpha P(X+1)+\beta Q(X+1),\qquad (\alpha P+\beta Q)(1)=\alpha P(1)+\beta Q(1),$$

we obtain

$$\begin{aligned} T_3(\alpha P+\beta Q) &= \alpha\big(P(X+1)-P(1)\big)\\ &\quad +\beta\big(Q(X+1)-Q(1)\big)\\ &= \alpha T_3(P)+\beta T_3(Q). \end{aligned}$$

Thus T_3 is linear.

MOTIVATION AND GUIDING IDEA

Two guiding questions. The kernel records information lost by a map; the image records attainable outputs. A trivial kernel means that distinct inputs remain distinguishable. An image equal to the codomain means that every output in that codomain is attainable.

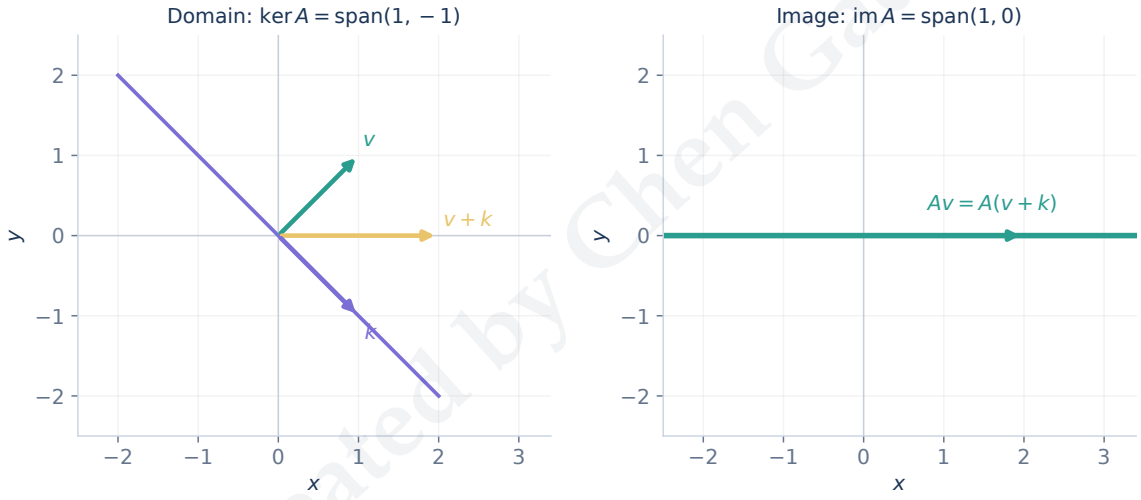


Figure 8.1. The kernel lies in the domain and the image in the codomain. Several inputs can have the same output when the kernel is nontrivial.

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8.2 The kernel and image of a linear map

RECALL AND TOOLS

Let $T: V \to W$ be linear.
The **kernel of** T consists of all vectors sent to 0_W:

$$\ker(T)=\{v\in V \mid T(v)=0\}$$

The kernel is a subspace of V.
The **image of** T is the set of all outputs attained by the map:

$$\mathrm{Im}(T)=\{T(v)\mid v\in V\}=\{w\in W\mid \exists v\in V, T(v)=w\}$$

The image is a subspace of W. It consists of outputs actually attained, whereas the codomain

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W is specified as part of the definition of the map. These need not be equal.
The sets $\ker T$ and $\operatorname{Im} T$ are nonempty, since $T(0_V) = 0_W$.

- Always $\operatorname{Im} T \subseteq W$. If $\operatorname{Im} T = W$, the map is **surjective**, or **onto**.
- T is **injective**, or **one-to-one**, if, for all $v_1, v_2 \in V$,

$$T(v_1) = T(v_2) \implies v_1 = v_2$$

  Equivalently, no two distinct inputs have the same image.

$$v_1 \neq v_2 \implies T(v_1) \neq T(v_2)$$

- A linear map T is an **isomorphism** if it is injective and surjective.
- Vector spaces V, W are **isomorphic** if there **exists** an isomorphism $T : V \to W$. We then write $V \cong W$.

PROPOSITION 8.1

A linear map $T : V \to W$ is **injective** if and only if $\ker T = \{0_V\}$.
For a homogeneous linear system, this means that the trivial solution is the only solution.

PROPOSITION 8.2

Let $T : V \to W$ be a linear map between finite-dimensional vector spaces. Then

1. If T is injective, $\dim V \leq \dim W$.
2. If T is surjective, $\dim W \leq \dim V$.
3. If $\dim V = \dim W$, then T is injective **if and only if** it is surjective.

EXERCISE 8.2

Let $n \geq 2$ and $V = M_n(\mathbb{F})$. Define $T : V \to \mathbb{F}$ by

$$T(A) = \operatorname{Tr}(A).$$

1. Please prove that T is linear.
2. Please find $\ker T$.
3. Please find $\operatorname{Im} T$.
4. Please compute $\dim \ker T$ and $\dim \operatorname{Im} T$, and please verify rank-nullity.

WORKED SOLUTION

Write $A = (a_{ij}), B = (b_{ij})$.

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EXERCISE 8.2 CONTINUED

1. Linearity:

$$T(A+B) = \mathrm{Tr}(A+B) = \sum_{i=1}^{n}(a_{ii}+b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii} = T(A)+T(B),$$

   For every $\lambda \in \mathbb{F}$,

$$T(\lambda A) = \mathrm{Tr}(\lambda A) = \sum_{i=1}^{n} \lambda a_{ii} = \lambda \sum_{i=1}^{n} a_{ii} = \lambda T(A).$$

   Thus T is linear.

2. Kernel:

$$\ker(T) = \{A \in M_n(\mathbb{F}) \ : \ \mathrm{Tr}(A) = 0\}.$$

   This is the subspace of matrices with trace zero.

3. Image: for any $\alpha \in \mathbb{F}$, take

$$A = \mathrm{diag}(\alpha, 0, \ldots, 0) \in M_n(\mathbb{F}),$$

   Then $\mathrm{Tr}(A) = \alpha$, so

$$\mathrm{Im}(T) = \mathbb{F}.$$

4. Dimensions: $\dim M_n(\mathbb{F}) = n^2$. Since $\mathrm{Im}\, T = \mathbb{F}$, its dimension is 1. Rank-nullity therefore gives

$$\dim \ker(T) = n^2 - 1.$$

   Alternatively, $n^2 - 1$ entries may be chosen freely, and the last diagonal entry is fixed by requiring trace zero.

EXERCISE 8.3

Let $V = C[0,1]$, the real vector space of continuous real-valued functions on $[0,1]$. Define

$$S : V \to \mathbb{R}, \qquad S(f) = \int_0^1 f(x)\,dx.$$

1. Please prove that S is linear.
2. Please find $\ker S$.
3. Please find $\mathrm{Im}\, S$.
4. Please give an explicit nonzero $f \in \ker S$.

WORKED SOLUTION

1. Linearity: for $f, g \in V$ and $\lambda \in \mathbb{R}$,

$$S(f+g) = \int_0^1 (f+g) = \int_0^1 f + \int_0^1 g = S(f) + S(g),$$

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EXERCISE 8.3 CONTINUED

$$S(\lambda f) = \int_0^1 \lambda f = \lambda \int_0^1 f = \lambda S(f).$$

so S is linear.

2. Kernel:
$$\ker(S) = \left\{ f \in C[0,1] \ : \ \int_0^1 f(x)\,dx = 0 \right\}.$$
Since $C[0,1]$ is infinite-dimensional, the finite-dimensional rank-nullity formula cannot here be treated as an equality of finite integers. Nevertheless,
$$C[0,1] = \ker(S) \oplus \operatorname{span}\{1\}, \qquad f = \big(f - S(f)\cdot 1\big) + S(f)\cdot 1.$$
The first summand has integral zero, and the only constant function with integral zero is the zero function.

3. Image: given $c \in \mathbb{R}$, take $f(x) \equiv c$. Then
$$S(f) = \int_0^1 c\,dx = c,$$
so $\operatorname{Im} S = \mathbb{R}$.

4. Example: $f(x) = x - \frac{1}{2}$ is continuous and nonzero, but
$$\int_0^1 \left(x - \tfrac{1}{2}\right) dx = \left[\tfrac{x^2}{2} - \tfrac{x}{2}\right]_0^1 = \tfrac{1}{2} - \tfrac{1}{2} = 0,$$
so $f \in \ker S$.

EXERCISE 8.4

Let V be a vector space over a field F, and let $T : V \to V$ be linear. Please prove:
(a) $\ker T \subseteq \ker T^2$; (b) $\operatorname{Im} T^2 \subseteq \operatorname{Im} T$.

WORKED SOLUTION

(a) We must show that every vector in the set on the left also belongs to the set on the right.
Let $v \in \ker T$. Then $T^2(v) = T(T(v)) = T(0) = 0$, so $v \in \ker T^2$.
(b) Let $w \in \operatorname{Im} T^2$. Some $v \in V$ satisfies $w = T^2(v) = T(T(v))$.
Set $u = T(v)$. Then $w = T(u)$, so $w \in \operatorname{Im} T$.

DEFINITION 8.3

If $T : V \to W$ is an **isomorphism**, each $w \in W$ has a unique $v \in V$ with $T(v) = w$.
Thus the **inverse map** $T^{-1} : W \to V$ is defined by $T^{-1}(w) = v$ whenever $T(v) = w$.
Applying T to v gives w; applying T^{-1} to w recovers v.

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EXERCISE 8.5

Let $V = M_3(\mathbb{R})$.

1. Please find a linear isomorphism $\Phi : V \to \mathbb{R}^9$.
2. In this part use the explicit row-by-row flattening isomorphism

$$\Phi((a_{ij})) = (a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{23}, a_{31}, a_{32}, a_{33}).$$

Define two subsets of $\mathbb{R}^9$:

$$U = \left\{ x \in \mathbb{R}^9 \;\middle|\; x_2 = x_4,\ x_3 = x_7,\ x_6 = x_8 \right\},$$

$$W = \left\{ x \in \mathbb{R}^9 \;\middle|\; x_1 = x_5 = x_9 = 0,\ x_2 = -x_4,\ x_3 = -x_7,\ x_6 = -x_8 \right\}.$$

Please find $\dim U, \dim W$ and please identify $A = \Phi^{-1}(U)$ and $B = \Phi^{-1}(W)$ as subspaces of $M_3(\mathbb{R})$.

WORKED SOLUTION

1. Define $\Phi : M_3(\mathbb{R}) \to \mathbb{R}^9$ by the following formula for $A = (a_{ij})$:

$$\Phi(A) = (a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{23}, a_{31}, a_{32}, a_{33}).$$

We verify that Φ is an isomorphism.

Linearity. Let $A = (a_{ij}), B = (b_{ij})$ and $\alpha, \beta \in \mathbb{R}$. The entries of $\alpha A + \beta B$ are $\alpha a_{ij} + \beta b_{ij}$, so

$$\Phi(\alpha A + \beta B) = (\alpha a_{11} + \beta b_{11}, \ldots, \alpha a_{33} + \beta b_{33}) = \alpha\Phi(A) + \beta\Phi(B).$$

Injectivity. If $\Phi(A) = \Phi(B)$, equality of all nine coordinates gives $a_{ij} = b_{ij}$ for all i, j, hence $A = B$.

Surjectivity. Given $x = (x_1, \ldots, x_9) \in \mathbb{R}^9$, define

$$A = \begin{pmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{pmatrix}.$$

Then $\Phi(A) = x$ by definition. Thus Φ is surjective.

Consequently Φ is an isomorphism.

2. Compute the dimensions and identify the corresponding matrix subspaces.

Computing $\dim U$**.** A vector $x \in U$ is determined by freely choosing

$$x_1,\ x_2,\ x_3,\ x_5,\ x_6,\ x_9,$$

The constraints then force

$$x_4 = x_2, \qquad x_7 = x_3, \qquad x_8 = x_6.$$

There are six free parameters, so

$$\dim U = 6.$$

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EXERCISE 8.5 CONTINUED

Computing $\dim W$**.** A vector $x \in W$ satisfies

$$x_1 = x_5 = x_9 = 0,$$

together with

$$x_4 = -x_2, \qquad x_7 = -x_3, \qquad x_8 = -x_6.$$

Exactly the following entries may be chosen freely:

$$x_2,\ x_3,\ x_6,$$

All remaining entries are then fixed. Hence

$$\dim W = 3.$$

Now identify $A = \Phi^{-1}(U)$. Let $A = (a_{ij}) \in M_3(\mathbb{R})$ and $\Phi(A) = (x_1, \ldots, x_9)$, so

$$\begin{aligned} x_1 &= a_{11}, \quad x_2 = a_{12}, \quad x_3 = a_{13}, \\ x_4 &= a_{21}, \quad x_5 = a_{22}, \quad x_6 = a_{23}, \\ x_7 &= a_{31}, \quad x_8 = a_{32}, \quad x_9 = a_{33}. \end{aligned}$$

The conditions defining U are

$$x_2 = x_4 \Rightarrow a_{12} = a_{21}, \qquad x_3 = x_7 \Rightarrow a_{13} = a_{31}, \qquad x_6 = x_8 \Rightarrow a_{23} = a_{32}.$$

These are precisely $A^T = A$, so

$$\Phi^{-1}(U) = \mathrm{Sym}_3 = \{A \in M_3(\mathbb{R}) \mid A^T = A\}.$$

Similarly, identify $B = \Phi^{-1}(W)$. The conditions defining W give

$$x_1 = x_5 = x_9 = 0 \Rightarrow a_{11} = a_{22} = a_{33} = 0,$$

and

$$x_2 = -x_4 \Rightarrow a_{12} = -a_{21}, \qquad x_3 = -x_7 \Rightarrow a_{13} = -a_{31}, \qquad x_6 = -x_8 \Rightarrow a_{23} = -a_{32}.$$

These are precisely $A^T = -A$, so

$$\Phi^{-1}(W) = \mathrm{Skew}_3 = \{A \in M_3(\mathbb{R}) \mid A^T = -A\}.$$

Thus:

$$\dim U = 6, \qquad \dim W = 3, \qquad \Phi^{-1}(U) = \mathrm{Sym}_3, \qquad \Phi^{-1}(W) = \mathrm{Skew}_3.$$

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8.3 Rank-nullity for linear maps

PROPOSITION 8.3

Let $T : V \to W$ be linear, with V finite-dimensional.
Then

$$\dim V = \dim \ker T + \dim \operatorname{Im} T.$$

EXERCISE 8.6

Please determine whether a linear map with the following properties exists. If it does, please give it explicitly.

1. $T : \mathbb{R}^3 \to \mathbb{R}^3$, with $\ker T = \operatorname{span}\{(1,2,1)\}$, $\operatorname{Im} T = \operatorname{Sp}\{(1,1,1)\}$.
2. $T : M_{2\times 2}(\mathbb{R}) \to \mathbb{R}^2$, with $\ker T = \operatorname{span}\left\{\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right\}$.

WORKED SOLUTION

1. Both the image and kernel are spanned by one nonzero vector, so each has dimension 1. Rank-nullity would give

$$3 = \dim(R^3) = \dim ImT + \dim kerT = 1 + 1 = 2$$

a contradiction. Thus no such linear map exists.

2. No such map exists. The stated kernel is spanned by one nonzero vector (a matrix), so $\dim \ker T = 1$. By rank-nullity,

$$\dim ImT = \dim(M_{2\times 2}(R)) - \dim \ker T = 4 - 1 = 3$$

But $\operatorname{Im} T \subseteq \mathbb{R}^2$, so $\dim \operatorname{Im} T \leq 2$, a contradiction. Hence the required map does not exist.

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8.4 Coordinates

DEFINITION 8.4

Let V be a vector space over a field $\mathbb{F}$ with ordered basis $B = (v_1, \ldots, v_n)$.
Every $v \in V$ has a unique representation as a linear combination of the basis vectors. Thus there are unique $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$ such that

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n$$

The coefficients $\alpha_1, \ldots, \alpha_n$ are the **coordinates of** v **relative to** B.

The column vector $\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in \mathbb{F}^n$ is the **coordinate vector of** v **relative to** B, denoted $[v]_B$.

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EXERCISE 8.7

Let $B = ((1,0,1)^T, (0,1,1)^T, (1,1,1)^T)$ be an ordered basis of $\mathbb{R}^3$. Please find the coordinates of $v = (5,2,4)^T$ relative to B, and please verify them by reconstructing v.

WORKED SOLUTION

Seek a, b, c such that $v = a(1,0,1)^T + b(0,1,1)^T + c(1,1,1)^T$. Comparing entries gives

$$a + c = 5, \qquad b + c = 2, \qquad a + b + c = 4.$$

Adding the first two equations and subtracting the third yields $c = 5 + 2 - 4 = 3$. Thus $a = 2$ and $b = -1$, so

$$[v]_B = (2, -1, 3)^T.$$

Check: $2(1,0,1)^T - (0,1,1)^T + 3(1,1,1)^T = (5,2,4)^T$. Uniqueness follows since the matrix of basis columns has determinant $-1 \neq 0$.

Conversely, knowing a vector's coordinates in a chosen basis allows us to reconstruct the vector.

With the notation above, $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ **if and only if** $[v]_B = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$.

EXERCISE 8.8

For the ordered basis $B = ((1,2)^T, (2,3)^T)$, suppose $[v]_B = (-2,3)^T$. Please find v in standard coordinates.

WORKED SOLUTION

The coordinate entries are the coefficients of the basis vectors:

$$v = -2\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 3\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -2+6 \\ -4+9 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}.$$

This is a basis since $\det \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = 3 - 4 = -1$. The column $(-2,3)^T$ gives coefficients relative to B, while $(4,5)^T$ gives coefficients relative to the standard basis. They represent the same vector.

PROPOSITION 8.4

Let V have dimension n over a field $\mathbb{F}$, and let B be an ordered basis of V.
The coordinate map $T_B : V \to \mathbb{F}^n$, defined by $T_B(v) = [v]_B$, is an isomorphism. In particular, $V \cong \mathbb{F}^n$.

RECALL AND TOOLS

Let V be a vector space over $\mathbb{F}$, let B be an ordered basis with n vectors, and let $u_1, \ldots, u_m$ be vectors in V. Then:

1. $\{u_1, \ldots, u_m\}$ is linearly independent (in V) **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ is linearly

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independent (in $\mathbb{F}^n$).

2. $\{u_1, \ldots, u_m\}$ spans V **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ spans $\mathbb{F}^n$.

3. $\{u_1, \ldots, u_m\}$ is a basis of V **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ is a basis of $\mathbb{F}^n$.

EXERCISE 8.9

Using coordinates in the ordered basis $(1, X, X^2)$, please show that the polynomials

$$p_1 = 1 + X, \qquad p_2 = X + X^2, \qquad p_3 = 1 + X + X^2$$

are linearly independent in $\mathbb{R}_2[X]$.

WORKED SOLUTION

Their coordinate columns are

$$[p_1] = (1, 1, 0)^T, \qquad [p_2] = (0, 1, 1)^T, \qquad [p_3] = (1, 1, 1)^T.$$

If $ap_1 + bp_2 + cp_3 = 0$, comparison of the coefficients of $1, X, X^2$ gives

$$a + c = 0, \qquad a + b + c = 0, \qquad b + c = 0.$$

Subtracting the first equation from the second gives $b = 0$. The third then gives $c = 0$, and the first gives $a = 0$. Thus only the trivial combination vanishes, proving independence. Since the space has dimension 3, they also form a basis; the coordinate matrix has determinant 1.

EXERCISE 8.10

Consider the real vector space $M_{2\times 2}(\mathbb{R})$.
Define two subspaces

$$U := \text{span}\{U_1, U_2, U_3\}, \qquad V := \text{span}\{V_1, V_2\},$$

where

$$U_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad U_3 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix},$$

$$V_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad V_2 = \begin{pmatrix} 0 & 1 \\ 2 & 2 \end{pmatrix}.$$

1. Please find a basis and the dimension of U and of V.

2. Please find a basis and the dimension of $U \cap V$.

WORKED SOLUTION

Use the linear map

$$\Phi : M_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4, \qquad \Phi\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = (a, b, c, d),$$

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EXERCISE 8.10 CONTINUED

which is bijective and hence preserves linear independence and dimensions.

1. Find bases and dimensions of U and V.

   Their coordinates are

   $$\Phi(U_1) = (1,0,0,0), \qquad \Phi(U_2) = (0,1,0,0), \qquad \Phi(U_3) = (0,0,1,1).$$

   If

   $$\alpha\Phi(U_1) + \beta\Phi(U_2) + \gamma\Phi(U_3) = (0,0,0,0),$$

   comparison of entries gives

   $$(\alpha,\beta,\gamma,\gamma) = (0,0,0,0) \Rightarrow \alpha = \beta = \gamma = 0.$$

   Thus $\{\Phi(U_1), \Phi(U_2), \Phi(U_3)\}$ is independent. Therefore $\{U_1, U_2, U_3\}$ is a basis of U, and

   $$\dim U = 3.$$

   For V,

   $$\Phi(V_1) = (1,0,1,1), \qquad \Phi(V_2) = (0,1,2,2).$$

   if

   $$\alpha\Phi(V_1) + \beta\Phi(V_2) = (0,0,0,0),$$

   we obtain

   $$(\alpha,\beta,\alpha+2\beta,\alpha+2\beta) = (0,0,0,0) \Rightarrow \alpha = 0,\ \beta = 0.$$

   Thus $\{\Phi(V_1), \Phi(V_2)\}$ is independent. Therefore $\{V_1, V_2\}$ is a basis of V, and

   $$\dim V = 2.$$

2. Find a basis and the dimension of $U \cap V$.

   First show that $V \subseteq U$ by expressing its generators in terms of the generators of U:

   $$V_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = U_1 + U_3,$$

   $$V_2 = \begin{pmatrix} 0 & 1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + 2\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = U_2 + 2U_3.$$

   Hence $V_1, V_2 \in U$, and $V \subseteq U$.

   Whenever $V \subseteq U$, we have $U \cap V = V$. Therefore,

   $$U \cap V = V = \operatorname{span}\{V_1, V_2\}, \qquad \dim(U \cap V) = \dim V = 2.$$

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8.5 Five descriptions of a linear map

Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ so that, for every

$$(x, y) \in \mathbb{R}^2$$

we have

$$T(x, y) = (x + y,\ x - y).$$

We now describe this same map in several ways.

Method 1: A formula for a general vector

This is a **general formula**: it specifies $T(v)$ for arbitrary $v = (x, y) \in \mathbb{R}^2$, rather than just particular vectors:

$$T(x, y) = (x + y,\ x - y).$$

It defines the map directly on its entire domain.

Method 2: Images of a basis

Choose the standard basis:

$$e_1 = (1, 0), \qquad e_2 = (0, 1).$$

Then

$$T(e_1) = T(1, 0) = (1, 1), \qquad T(e_2) = T(0, 1) = (1, -1).$$

For $(x, y) = xe_1 + ye_2$, extend linearly:

$$T(x, y) = xT(e_1) + yT(e_2) = x(1, 1) + y(1, -1) = (x + y,\ x - y).$$

Thus specifying the images of a basis defines the same map.

Method 3: A matrix

Write the input as a column:

$$\begin{pmatrix} x \\ y \end{pmatrix}.$$

Define

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$T(x, y) = A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ x - y \end{pmatrix}.$$

This matrix represents the same map in the standard basis.

Method 4: Familiar linear operations

View T as performing the following operations:

- First component: sum of the coordinates.
- Second component: difference of the coordinates.

Thus

$$T(x, y) = \big(\text{sum}(x, y), \text{diff}(x, y)\big),$$

These operations are linear, so T is linear.

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Method 5: A structural decomposition

Specify that T satisfies

$$T(e_1 + e_2) = (2, 0), \qquad T(e_1 - e_2) = (0, 2),$$

and that T is linear.

Since $\{e_1+e_2, e_1-e_2\}$ is a basis of $\mathbb{R}^2$, these requirements determine T uniquely. Calculation gives the same formula:

$$T(x, y) = (x + y,\ x - y).$$

Remarks

- A **general formula** defines $T(v)$ for an arbitrary vector v.
- A linear map may be specified by a formula, basis images, a matrix or a structural description.
- The descriptions differ, but the map itself is the same.

8.5.1 Summary of the methods

RECALL AND TOOLS

1. **An explicit formula for a general vector**
   Specify $T(v)$ for every $v \in V$.
   Check: the formula is well defined and satisfies both linearity conditions.

2. **Basis images and linear extension**
   If $B = \{v_1, \dots, v_n\}$ is a basis of V, freely choose

   $$T(v_1), \dots, T(v_n) \in W$$

   and define, for $v = \sum_{i=1}^{n} \alpha_i v_i$,

   $$T(v) := \sum_{i=1}^{n} \alpha_i\, T(v_i).$$

   Each choice of basis images determines exactly one linear map.

3. **A matrix**
   For $V = \mathbb{F}^n$ and $W = \mathbb{F}^m$, define

   $$T(x) = Ax$$

   where A is an $m \times n$ matrix.
   Every matrix defines a linear map. After choosing ordered bases, every linear map between finite-dimensional spaces has a matrix representation.
   Remark: the matrix depends on the chosen bases; the map itself does not.

4. **A familiar linear operation**
   Use an operation such as differentiation, integration or multiplication by a fixed scalar.
   Examples:

   $$T(P) = P', \qquad T(f) = \int_0^1 f(x)\, dx.$$

   Specify a domain and codomain, check well-definedness and verify linearity in each

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case.

5. **A direct-sum decomposition**
   A map can be specified by its action on the components of a direct sum.
   For example, if $V = U \oplus W$, define T by

$$T(u) = 0 \ \ (\forall u \in U), \qquad T(w) = w \ \ (\forall w \in W).$$

   Linearity follows from the direct-sum decomposition and the linear action on each component.

Common errors

- Adding a nonzero constant vector destroys linearity: $T(v) = v + b$ with $b \neq 0$ is not linear.
- Always specify the domain and codomain: $T : V \to W$. The image $\operatorname{Im} T$ may be a proper subspace of the codomain.
- Distinguish a vector, its coordinate vector in an ordered basis, and the matrix representing a map.

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EXERCISE 8.11

A system receives three samples $x = (x_1, x_2, x_3)^T$ and subtracts their mean. Define

$$P(x) = x - \frac{x_1 + x_2 + x_3}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Please prove that P is linear, please find its kernel and image, and please show that a second application changes nothing: $P^2 = P$.

WORKED SOLUTION

Write $\mu(x) = (x_1 + x_2 + x_3)/3$ and $\mathbf{1} = (1, 1, 1)^T$. For vectors x, y and scalars a, b,

$$\mu(ax + by) = a\mu(x) + b\mu(y),$$

so

$$P(ax + by) = ax + by - \big(a\mu(x) + b\mu(y)\big)\mathbf{1} = aP(x) + bP(y).$$

Thus P is linear. If $P(x) = 0$, then $x = \mu(x)\mathbf{1}$; conversely, every constant vector is sent to zero. Therefore,

$$\ker P = \operatorname{span}\{\mathbf{1}\}, \qquad \dim \ker P = 1.$$

For every x, the sum of the entries of $P(x)$ is $x_1 + x_2 + x_3 - 3\mu(x) = 0$. Hence its image lies in the plane

$$H = \{y \in \mathbb{R}^3 : y_1 + y_2 + y_3 = 0\}.$$

If $y \in H$, then $\mu(y) = 0$ and $P(y) = y$. Thus $H \subseteq \operatorname{Im} P$, giving

$$\operatorname{Im} P = H = \operatorname{span}\{(1, -1, 0)^T, (1, 0, -1)^T\}, \qquad \dim \operatorname{Im} P = 2.$$

Finally, $P(x) \in H$, so $P(P(x)) = P(x)$. The dimensions satisfy $1 + 2 = 3$. Removing the mean eliminates the constant component and retains the differences between samples. After

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EXERCISE 8.11 CONTINUED

one application the mean is already zero.

Additional practice

EXERCISE 8.12 FOUNDATION

Define $T : \mathbb{R}_{\leq 2}[X] \to \mathbb{R}^2$ by

$$T(p) = \begin{pmatrix} p(1) \\ p(-1) \end{pmatrix}.$$

Please prove linearity, please find the kernel and image, and please verify rank-nullity. Please construct a linear map $S : \mathbb{R}^2 \to \mathbb{R}_{\leq 2}[X]$ with $T \circ S = I_{\mathbb{R}^2}$. Is $S \circ T$ also the identity?

WORKED SOLUTION

Evaluation preserves addition and scalar multiplication, so T is linear. For $p = a + bX + cX^2$, membership in the kernel requires

$$a + b + c = 0, \qquad a - b + c = 0.$$

Subtracting gives $b = 0$, hence $a = -c$. Thus $\ker T = \operatorname{span}\{X^2 - 1\}$ has dimension 1. For every $(u, v)^T \in \mathbb{R}^2$, define

$$S((u, v)^T) = \frac{u + v}{2} + \frac{u - v}{2} X.$$

This polynomial has values u, v at $1, -1$, respectively, and its coefficients are linear in u, v. Hence S is linear, $T \circ S = I_{\mathbb{R}^2}$ and $\operatorname{Im} T = \mathbb{R}^2$. Indeed, $1 + 2 = 3 = \dim \mathbb{R}_{\leq 2}[X]$. However,

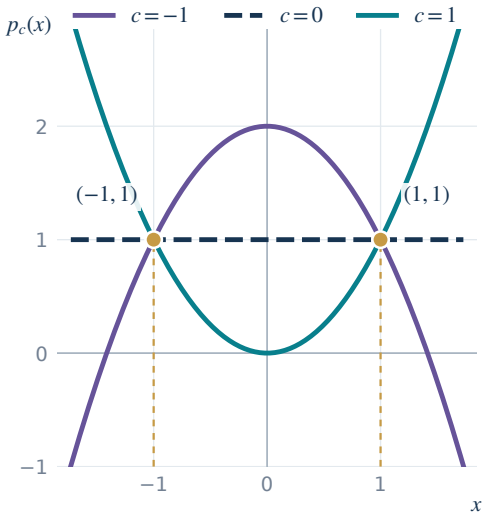
$$(S \circ T)(X^2) = S((1, 1)^T) = 1 \neq X^2,$$

so $S \circ T$ is not the identity. Explicitly, $(S \circ T)(a + bX + cX^2) = a + c + bX$: the measurements do not determine the quadratic coefficient.

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$$p_c(x) = 1 + c(x^2 - 1), \qquad c \in \{-1, 0, 1\}$$



$$T(p) = (p(1), p(-1))^T$$

$$p_{-1}(x) = 2 - x^2, \quad p_0(x) = 1, \quad p_1(x) = x^2$$

$$p_c(-1) = p_c(1) = 1$$

$$T(p_{-1}) = T(p_0) = T(p_1) = (1, 1)^T$$

$$\ker T = \operatorname{span}\{X^2 - 1\}$$

$$p_c - p_0 = c(X^2 - 1)$$

Figure 8.2. Three distinct polynomials give the same two measurements: $p_c(1) = p_c(-1) = 1$ for $c = -1, 0, 1$. Their differences $p_c - p_0 = c(X^2 - 1)$ lie in the kernel of $T(p) = (p(1), p(-1))^T$, so these measurements do not distinguish the polynomials.

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EXERCISE 8.13 CHALLENGE

Let $P : V \to V$ be linear and satisfy $P^2 = P$. Please prove

$$V = \ker P \oplus \operatorname{Im} P, \qquad \ker(I_V - P) = \operatorname{Im} P, \qquad \operatorname{Im}(I_V - P) = \ker P.$$

Please prove that every specified decomposition $V = U \oplus W$ determines a unique such map with kernel U and image W. Please find both components of the decomposition for

$$P(x, y, z) = (x - z, y + 2z, 0) \quad \text{on } \mathbb{R}^3.$$

WORKED SOLUTION

For every $v \in V$, write $v = (v - Pv) + Pv$. The second term is in the image, and the first is in the kernel because $P(v - Pv) = Pv - P^2v = 0$. If $w = Pv \in \operatorname{Im} P$, then $Pw = P^2v = Pv = w$. Thus a vector in both kernel and image must be zero, proving the direct sum. Also, $(I_V - P)w = 0$ is equivalent to $Pw = w$, and hence to $w \in \operatorname{Im} P$. The decomposition proves $\operatorname{Im}(I_V - P) \subseteq \ker P$. Conversely, if $u \in \ker P$, then $(I_V - P)u = u$. For $V = U \oplus W$, define $P(u + w) = w$ using the unique decomposition. Uniqueness makes the map well defined and linear. Immediately $P^2 = P$, its kernel is U and its image is W. Every map with these properties must annihilate U and act identically on W, proving uniqueness. In the example,

$$\ker P = \operatorname{span}\{(1, -2, 1)^T\}, \qquad \operatorname{Im} P = \{(a, b, 0)^T : a, b \in \mathbb{R}\},$$

and the explicit decomposition is

$$(x, y, z)^T = z(1, -2, 1)^T + (x - z, y + 2z, 0)^T.$$

Applying P again fixes the second component, so $P^2 = P$. This is projection along a direct-sum complement; orthogonality is not assumed.

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EXERCISE 8.14 CHALLENGE

Let V have finite dimension $n \geq 2$ over a field F. Let $T : V \to V$ be linear and satisfy $T(v) \in \text{span}\{v\}$ for every $v \in V$. Please prove that one scalar λ satisfies $T(v) = \lambda v$ for all v. Please check whether imposing the condition only on a fixed basis is sufficient.

WORKED SOLUTION

Choose a basis $e_1, \ldots, e_n$ and write $T(e_j) = \lambda_j e_j$. For $i \neq j$, the hypothesis for $e_i + e_j$ gives

$$\lambda_i e_i + \lambda_j e_j = T(e_i + e_j) = \mu(e_i + e_j)$$

for some μ. Independence implies $\lambda_i = \lambda_j = \mu$. All basis scalars therefore equal one λ, and linearity gives

$$T\left(\sum_j a_j e_j\right) = \sum_j a_j \lambda e_j = \lambda \sum_j a_j e_j.$$

A basis condition alone is insufficient. On $\mathbb{R}^2$, $T(x, y) = (x, 2y)$ preserves the lines of the two standard basis vectors, but $T(1, 1) = (1, 2)$ is not a multiple of $(1, 1)$.

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EXERCISE 8.15 CHALLENGE

Let V, W, U be finite-dimensional spaces over a field F, and let $S : V \to W$ and $T : V \to U$ be linear. Please prove that $\ker S \subseteq \ker T$ if and only if there is a linear map $R : W \to U$ with $T = R \circ S$. When S is onto, please prove that R is unique.

WORKED SOLUTION

If $T = R \circ S$ and $x \in \ker S$, then $T(x) = R(0) = 0$, so $x \in \ker T$. Conversely, assume the kernel inclusion and define on $\operatorname{Im} S$

$$R_0(Sx) = Tx.$$

This is well-defined: if $Sx = Sy$, then $x - y \in \ker S \subseteq \ker T$, so $Tx = Ty$. Also

$$R_0(aSx + bSy) = R_0(S(ax + by)) = T(ax + by) = aTx + bTy,$$

which proves linearity. Choose a basis of $\operatorname{Im} S$ and extend it to a basis of W. Define R by R_0 on the first basis vectors and by zero on the added vectors, then extend linearly. This gives $R(Sx) = Tx$ for every x. If S is onto, every $w \in W$ is Sx for some x. The equality forces $R(w) = Tx$, so there is only one possible map R.

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